

KEY FORMULA

Symbols

Symbol(s)	Meaning	Value / Notes
Architecture (fixed per model)		
L, L_{moe}	transformer layers; layers carrying MoE	43/43, 61/61, 61/58
n_{csa}, n_{hca}	CSA / HCA layer counts	from compress_ratios
e, ι	bytes per KV entry; per CSA indexer key	576, 68
m, m'	CSA / HCA compression ratios	4, 128
n_{vin}, k_{top}	sliding window; CSA sparse top-k	128; 512/1024
n_e, k	routed experts per layer; activated per token	256/384; 6/8
w_e, W_0	parameters per expert; non-expert weight bytes	
b_e, b	bytes per expert param; per stored KV element	0.53125 (MXFP4) / 1.0
P_{act}, c_{max}	activated params per token; context ceiling	
d_{kv}, d_r	MLA latent / decoupled-RoPE dims (dense case)	512, 64
Workload, persona i		
s_i, c_i, o_i	seat share; context tokens; output tokens	
r_i, h_i	requests per active user per day; prefix-cache hit rate	
λ_i, C_i	peak arrival rate; concurrent sessions	
Deployment		

Symbol(s)	Meaning	Value / Notes
N, a, φ	seats; daily adoption; peak-hour concentration	
ℓ , ψ	planned load factor; idle-KV multiplier	0.70, 1.0
τ , T_0	TPOT SLO (s/token); TTFT budget (s)	0.040, 2.0
η_m , η_β , η_{mfu}	usable HBM; achievable bandwidth; prefill MFU	0.85, 0.70, 0.40
σ , R, v	expert routing skew; redundancy; GPUs per node	1.15, 2, 8
Hardware		
H, β , F	HBM bytes; HBM bandwidth; dense FP8 FLOP/s	

Formula

Stage 0 — Model constants (deployment-independent)

$\kappa = n_{\text{csa}} \frac{e + \iota}{m} + n_{\text{hca}} \frac{e}{m'}$	KV bytes <i>stored</i> per context token
$A = e (n_{\text{csa}} k_{\text{top}} + L n_{\text{win}})$	KV bytes <i>read</i> per step, context-free
$B = n_{\text{csa}} \frac{\iota}{m} + n_{\text{hca}} \frac{e}{m'}$	KV bytes <i>read</i> per step, per context token
$\rho(c) = A + Bc$	KV read per sequence per decode step
$W = L_{\text{moe}} (n_e + 1) w_e b_e + W_0$	total weight bytes

Stage 1 — Demand to concurrency (Little’s Law)

$$\lambda_i = \frac{N s_i a r_i \varphi}{3600}$$

$$\gamma_i = \frac{a \varphi}{3600 \ell} s_i r_i (T_0 + o_i \tau), \quad \gamma = \sum_i \gamma_i, \quad C_i = \gamma_i N, \quad C = \gamma N$$

Stage 2 — Capacity

$$M = W + \psi \kappa \sum_i C_i c_i = W + \psi \kappa \bar{c} \gamma N, \quad G_{\text{cap}} = \frac{M}{H \eta_m}$$

Stage 3 — Decode bandwidth

$$W_{\text{eff}}(C) = L_{\text{moe}} \left(n_e \min \left[1, \sigma \left(1 - \left(1 - \frac{k}{n_e} \right)^C \right) \right] + 1 \right) w_e b_e + W_0$$

$$S(C) = W_{\text{eff}}(C) + A C + B \sum_i C_i c_i, \quad G_{\text{bw}} = \frac{S(C)}{\beta \eta_b \tau}$$

Stage 3b — Prefill

$$\Pi = \sum_i \lambda_i c_i (1 - h_i) \quad \text{uncached prompt tokens/s at peak}$$

$$G_{\text{pf}} = \frac{2 \Pi P_{\text{act}}}{F \eta_{\text{mfu}}} \quad \text{prefill GPUs}$$

Stage 4 — Feasibility and fleet

$$\text{Infeasible if } \max_i c_i > c_{\text{max}}$$

$$\mathcal{S}(g) = \begin{cases} \min\{x \in \{1, 2, 4, \nu\} : x \geq g\}, & g \leq \nu \\ \nu \lceil g/\nu \rceil, & g > \nu \end{cases} \quad (\text{snap to a valid TP/EP degree})$$

$$G = R[\mathcal{S}(\max(G_{\text{cap}}, G_{\text{bw}})) + \mathcal{S}(G_{\text{pf}})]$$

Derived invariants

$$\Lambda = \frac{W}{\kappa} \quad \text{live-context tokens whose KV equals } W$$

$$\Phi = \frac{B}{\kappa} \quad \text{re-read fraction (dense} = 1)$$

$$C_{\text{sat}} = \frac{\ln(1 - \varepsilon)}{\ln(1 - k/n_e)} \quad \text{batch at which } W_{\text{eff}} \rightarrow \varepsilon W$$

$$N^* = \frac{\Lambda}{\psi \gamma \bar{c}} \quad \text{headcount where KV memory} = W$$

The latex formulas are rendered by google gemini. Not produced by it.

UNCERTAINTY REGISTER

What this register is for:

Explains **which numbers are derived and verified, which are absorbed into the tool as parameters, and which are still guesses.**

Key disclaimer:

Model constants were derived from architecture and confirmed field-by-field against config.json; workload inputs were invented. Those deserve very different levels of trust and the fleet number alone doesn't distinguish them.

Broadly the rows fall into three classes:

Architectural facts (U1) are binary and resolved by a procurement decision, not by more analysis.
Derived model constants (U2, U3, U5, U6) are narrow and verified; collectively they move the fleet by roughly one GPU.

Workload and efficiency assumptions (U4, U7–U11) are wide, unmeasured, and can only be closed by a pilot.

#	Source	Affects	Magnitude	Modelled?	Character
U1	FP4 hardware path	W	+84% Flash, +86% Pro (+727 GB on Pro)	Yes — HW[].fp4_native	Binary. A procurement decision, not an estimate. Dominates all other rows combined.
U8	GPU FLOPS figures unverified	G_{pf}	Unknown; prefill is ~50% of the Hopper fleet	No	Values in HW[] are from memory. Verify against vendor datasheets before spending.
U9	Prefix-cache hit rates assumed (0.30 / 0.85 / 0.20)	G_{pf}	High leverage — sets Π directly	Input only	Measurable in a pilot; currently pure assumption.

#	Source	Affects	Magnitude	Modelled?	Character
U4	Expert routing skew	G_{bw}	+10–20%	Yes — $\sigma = 1.15$	Absorbed, but σ itself is unvalidated. Stragglers set step time under EP.
U7	Arrival burstiness	C	Tail exceeds γN	Partly — $\ell = 0.70$	φ and ℓ are crude proxies for a real queueing model.
U10	Prefill MFU = 0.40	G_{pf}	$\pm 25\%$ plausible	Yes — constant	Absorbed as a fixed parameter; not validated on this architecture.
U11	Interconnect / EP network limits	G_{bw}	May bind at ≥ 64 GPUs	No	Tech report gives a 6,144 FLOPs/byte balance point for Pro; unmodelled.
U3	MTP module loaded	W	+3.5 / +13.8 / +11.5 GB	Yes — <code>--mtp</code>	Optional. Zero κ impact (compress_ratio: 0 confirmed for both MTP blocks).
U5	V4-Flash config provenance	Flash constants	Low	No	Third-party FlagOS mirror; every field cross-checks against the tech report.
U2	Indexer keys persisted?	κ	–11.5% / –11.4% on κ ; 0 GPUs on output	Yes — <code>--band</code>	Diagnosed: removing the indexer term reproduces the paper's figures to <0.5% on both models. Immaterial to the fleet.
U6	Embedding precision	W	± 0.5 / ± 0.9 GB	No	Negligible.

USING sizing_v2.py :

```
python3 sizing_v2.py          # full report, default assumptions
python3 sizing_v2.py --seats 12000 # your headcount
python3 sizing_v2.py --sweep    # scaling curve 500→50,000 seats; B200 only, ignores --seats
python3 sizing_v2.py --band lo  # optimistic KV accounting
python3 sizing_v2.py --mtp      # include MTP for speculative decoding
```

The CLI covers four common overrides. Everything else — SLOs, efficiency factors, redundancy, the workload mix, and hardware — is edited in the DEPLOY, MIX, and HW blocks at the foot of the file. Requires Python 3.8+, standard library only. Model constants self-validate on import and raise `AssertionError` if edited inconsistently.